Mikhail Kuznetsov

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Summary

Research Scientist specializing in securing agentic AI systems and foundation models for audit/security logs. Current focus: detecting intent deviation and policy violations in agent runtime traces, extending detections with OS-level signals (eBPF), and auto-detecting agentic workloads in dynamic environments. Led an audit-log FM end-to-end: diversity-preserving 10B → 10M training-data sampling, pretraining + contrastive fine-tuning, interpretable anomaly scoring, and evaluation with limited feedback. Ongoing research in LLM alignment and robust self-supervised learning.

Research Interests

Agentic AI security • Foundation models for logs • Robust/self-supervised learning • Anomaly detection • Representation learning for structured/tabular data • Evaluation & interpretability • LLM alignment

Experience

Amazon AWS – AI Security and Observability *2022 – Present*
Senior Applied Scientist

- **Agentic AI Security**: Leading detection for securing AI agents in production. Models over agent runtime traces (e.g. Amazon Bedrock AgentCore) detect intent deviation and policy violations — prompt injection being one driver — conditioned on an agent's historical behavior and agentic context (user/system prompts). Extending detections with OS-level telemetry (eBPF, system logs) to verify what an agent actually did against what its trace claims, and to auto-detection of agentic workloads in dynamic environments.
- **LLM-Based Foundation Model on Audit Logs**: Tech lead of a foundation model for audit logs, learning embeddings for security entities (IP, API, usernames) from dynamic audit data. The model was deployed to production in Sep 2025, enhancing Customer-facing anomaly detection, reducing False Positives by 20–30%. Main contributions are:

  Training data creation. Designed a sampling algorithm that compresses ∼10B raw log events to ∼10M high-value examples while preserving entity diversity; runs on a regular cadence to continually refresh training data.

  Contrastive Fine-tuning. Introduced contrastive fine-tuning for tenant-specific contexts, which are focused on entities particularly important for downstream consumers (asnOrg, api).

  Evaluation & interpretability. Built a multi-signal evaluation framework (weak supervision, third-party feeds, expert labels) and interpretability tooling (embedding similarity, contextual neighbors) for explainable anomalies.

  Architectural design (hierarchical logs). Co-designed a memory-efficient architecture leveraging the nested structure of log entries for long-context, high-throughput security workload.

  GenAI in Security Benchmarking: Co-authored a benchmark evaluating generative models in security workflows.
- **Robust/Secure Learning**: Co-developed distributionally robust self-supervised learning for tabular data; advanced graph-based domain reputation with weak supervision; ongoing research on agentic AI security for context-aware anomaly/failure detection in multi-agent systems.

Yahoo! Research, New York *2017 – 2021*
Research Scientist

- **ExtremeText**: contributed to a no-regret generalization of hierarchical softmax for *extreme* multi-label classification; improved training/inference efficiency and accuracy over very large label spaces.

- **VisualTextRank**: designed an unsupervised, graph-based content extraction method to automate ad text → image matching; formulated a TextRank-style ranking over multimodal (text/vision) features, built data/eval pipelines, and improved retrieval relevance in internal benchmarks.
- Built unified multi-task CTR/CVR models that improved advertising ROI by >10% in online A/B tests.

Education

Moscow Institute of Physics and Technology
Ph.D. (Computer Science, Mathematical Modeling), 2016
M.S. with Honors, Applied Mathematics & Physics, 2013
B.S. with Honors, Applied Mathematics & Physics, 2011

Yandex School of Data Analysis — Diploma in Data Science, 2012

Publications by Topic

AI Security & Observability

- *EV-AUDIT: A Co-Evolutionary Auditing Framework for Task Hijacking in Multi-Agent Systems.* T. Bilot, Z. Zhang, K. Liu, **M. Kuznetsov**, W. Ding. In submission, 2026.
- *HLogformer: A Hierarchical Transformer for Representing Log Data.* Z. Hou, M. Ghashami, **M. Kuznetsov**, MA. Torkamani. 2024
- *An Evaluation Benchmark for Generative AI in Security Domain.* M. Ghashami, **M. Kuznetsov**, V. Gao, G. Teng, P. Wallis, J. Xie, A. Torkamani, B. Coskun, W. Ding. KDD GenAI Evaluation Workshop, 2024.

Robust / Extreme Classification

- *Distributionally Robust Self-Supervised Learning for Tabular Data.* S. Ghosh, J. Xie, **M. Kuznetsov**. NeurIPS Third Table Representation Learning Workshop, 2024.
- *A No-Regret Generalization of Hierarchical Softmax to Extreme Multi-Label Classification.* M. Wydmuch, K. Jasinska, **M. Kuznetsov**, R. Busa-Fekete, K. Dembczynski. NeurIPS, 2018.

Language Model Alignment

- *STARS: Synchronous Token Alignment for Robust Supervision in Large Language Models.* M. Quamar, M. Areeb, **M. Kuznetsov**, M. Ozmen, B. Celik. SPIGM Workshop, ICML, 2026.
- *Adaptive Blockwise Search: Inference-Time Alignment for Large Language Models.* M. Quamar, M. Areeb, N. Sharma, Nishant, A. Shreekumar, J. Jonathan, M. Ozmen, **M. Kuznetsov**, B. Celik. 2025.

Generative Models & Retrieval for Ads/Vision

- *Salient Object-Aware Background Generation using Text-Guided Diffusion Models.* A. Eshratifar, J. Soares, K. Thadani, S. Mishra, **M. Kuznetsov**, Y. Ku, P. De Juan. CVPR Workshop on Generative Models for Computer Vision, 2024.
- *Staging E-Commerce Products for Online Advertising using Retrieval Assisted Image Generation.* Y. N. Ku, **M. Kuznetsov**, S. Mishra, P. de Juan. AdKDD, 2023.
- *VisualTextRank: Unsupervised Graph-Based Content Extraction for Automating Ad Text-to-Image Search.* KDD (ADS Track). S. Mishra, **M. Kuznetsov**, G. Srivastava, M. Sviridenko, 2021.

Open-Source Projects

ClawGuard — AI workload observer: auto-detects and watches agentic workloads on a machine (processes, network, parent chains) via an LLM loop that maintains a live world model in place of hand-written rules. github.com/mikkuzne/clawguard

apparty — Telegram bot that builds sandboxed (`bwrap`) web apps on demand with an Anthropic tool-use agent; a second model narrates the agent's actions in plain English. github.com/mikkuzne/apparty

pitchclaw — LLM-curated calibrated priors for football match outcomes: a weekly-rewritten team-strength model feeding a mechanical decision filter. github.com/mikkuzne/pitchclaw

Skills

Data & Infra: PySpark; Pandas; Arrow/Parquet; large-scale data curation/sampling; AWS (SageMaker, S3, EMR, EKS).

Modeling: scikit-learn; PyTorch; Hugging Face (Transformers, Accelerate, Datasets, PEFT); vLLM; LoRA/QLoRA.